

CONFLUENT CLOUD · REAL-TIME FRAUD DETECTION

Scam Shield

A live walkthrough of the running platform — stream lineage, schemas, connectors, Flink, the Terraform that builds it, and the agentic ops dashboard on top.

Stream Lineage

Schemas

Connectors

Flink

Terraform

Agentic AI

Confluent Kafka · Flink SQL · Schema Registry · Tableflow → Apache Iceberg · HashiCorp Terraform · on Microsoft Azure (East US)

What this walkthrough covers

Each slide pairs a live console screenshot with a short, self-contained note — following the platform from the streams up to the agentic dashboard.

01

Platform

One account, dedicated cluster, Private Link.

02

Stream Lineage

The pipeline mapped end to end, live.

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Schemas & Governance

Avro contracts; Advanced Stream Governance.

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Connectors & Tableflow

Managed sinks; zero-ETL to Iceberg.

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Flink

Nine SQL statements + governed AI narration.

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Terraform

Two roots; plan → apply through CI.

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Agentic Dashboard

Command center, HITL, learning loop.

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Cost

Itemised, per-service consumption & metering.

PLATFORM

One account, one private cluster

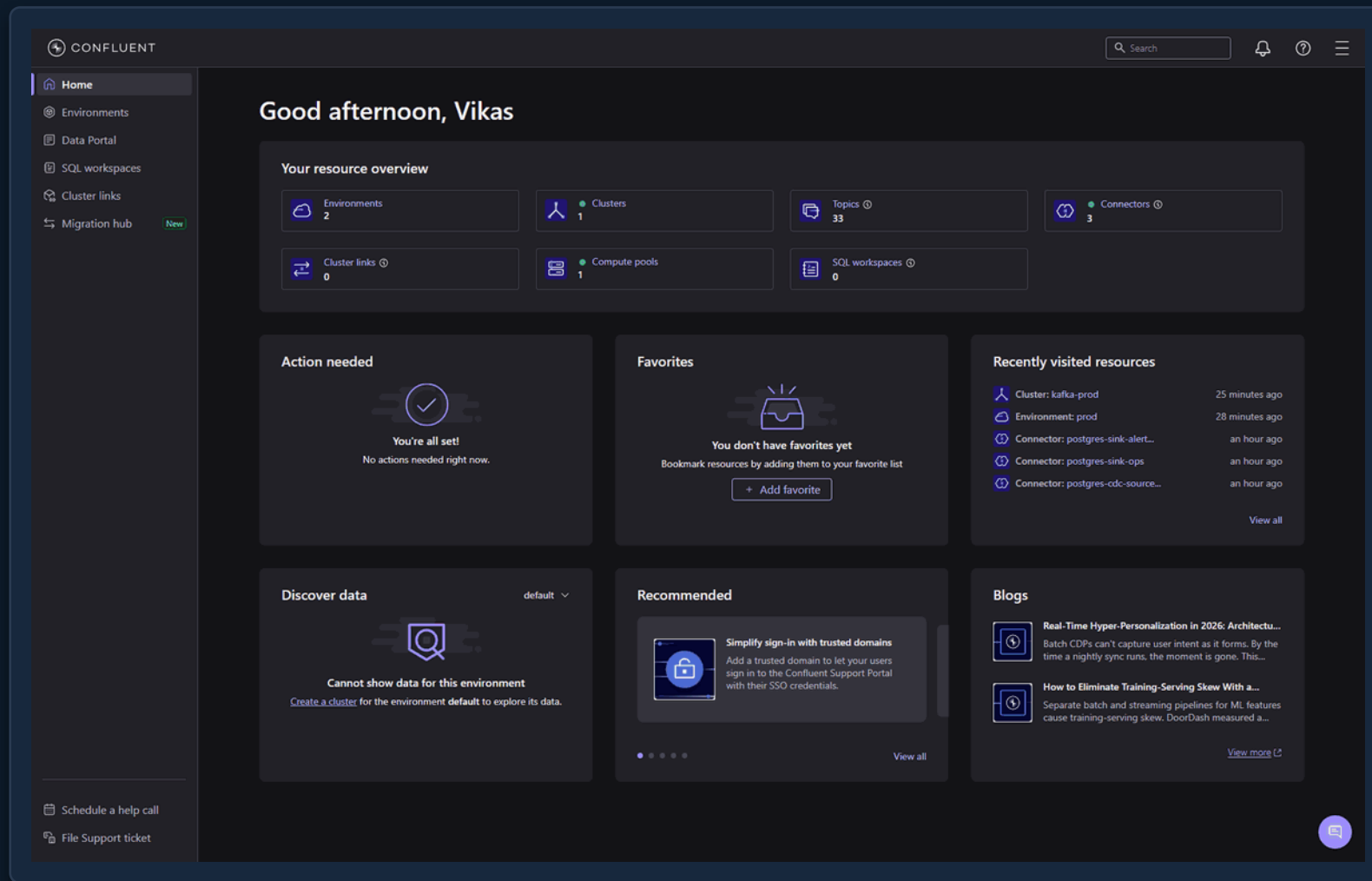
The production estate at a glance — a single dedicated Kafka cluster on Azure behind Private Link.

One Confluent account runs the platform

WHAT YOU'RE SEEING

- Production org: 1 environment, 1 Kafka cluster
- 32 topics, 3 connectors, 1 Flink compute pool
- Whole estate managed from one console
- Resource overview at a glance

WHY IT MATTERS *Kafka is the single source of truth for the entire fraud pipeline.*

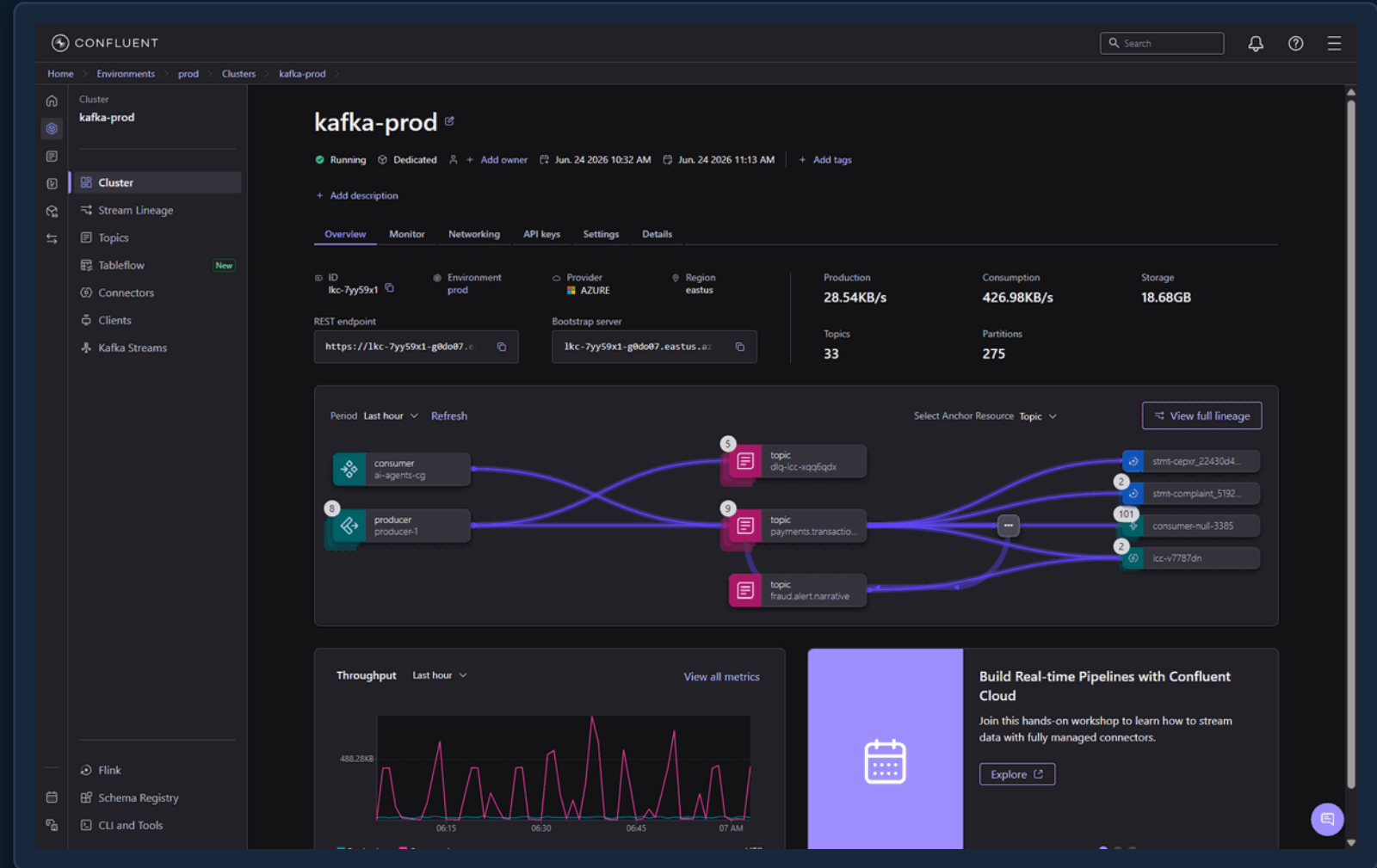


A dedicated, private cluster on Azure

WHAT YOU'RE SEEING

- kafka-prod — Dedicated, status Running
- Azure · eastus · 274 partitions · ~4.2 GB
- Live throughput and stream-lineage preview
- Enterprise SLA on private networking

WHY IT MATTERS *An isolated, enterprise-grade backbone — not a shared demo cluster.*



Private Link, no public exposure

WHAT YOU'RE SEEING

- ccn-confluent-prod private network
- Ingress / egress connections enumerated
- Private Link data plane
- No broker reachable over the internet

WHY IT MATTERS *The data plane is private by default — a zero-trust posture.*

The screenshot displays the Confluent Network Management interface for the 'prod' environment. The left sidebar shows navigation options: Environment overview, Clusters, Flink, Streaming agents, Network management (selected), Stream Governance, Schema Registry, Catalog management, Integrations, and Artifacts. The main panel shows the 'ccn-confluent-prod' network overview, which is a Private Link network. The configuration details are as follows:

Network overview	
ID	n-g0do07
Provider	Azure
Region	eastus
Connection types	Private Link
Confluent Cloud VNet	s-plz81
Confluent Cloud Subscription ID	a00af2b3-3548-4731-a2d8-7c063bf38b79
DNS resolution	Chased Private
DNS domain	g0do07.eastus.azure.confluent.cloud
Endpoint suffix	<service-identifier>-g0do07.eastus.azure.glb.confluent.cloud
Availability zone 1	az1.g0do07.eastus.azure.confluent.cloud
• DNS subdomain	s-plz81-privatelink-1.c5df8881-e64c-49ef-9010-6bc2cc781893.eastus.azure.privatelinkservice/subscriptions/a00af2b3-3548-4731-a2d8-7c063bf38b79/resourceGroups/s-plz81/providers/Microsoft.Network/privateLinkServices/s-plz81-privatelink-1
• Service Alias	
• Resource ID	
Availability zone 2	az2.g0do07.eastus.azure.confluent.cloud
• DNS subdomain	s-plz81-privatelink-2.ee507fe9-063c-48d6-9b94-aaf94fac4809.eastus.azure.privatelinkservice/subscriptions/a00af2b3-3548-4731-a2d8-7c063bf38b79/resourceGroups/s-plz81/providers/Microsoft.Network/privateLinkServices/s-plz81-privatelink-2
• Service Alias	
• Resource ID	
Availability zone 3	az3.g0do07.eastus.azure.confluent.cloud
• DNS subdomain	s-plz81-privatelink-3.b5bb5917-06dd-4975-a2d4-3d313cf0300a.eastus.azure.privatelinkservice/subscriptions/a00af2b3-3548-4731-a2d8-7c063bf38b79/resourceGroups/s-plz81/providers/Microsoft.Network/privateLinkServices/s-plz81-privatelink-3
• Service Alias	
• Resource ID	

Below the network overview, there are tabs for Clusters, Ingress connections (selected), Egress connections, and Egress DNS. The Ingress connections tab shows a search bar and a table with one entry, 'pla-confluent-prod'. A '+ Private Link Access' button is visible in the top right corner of the interface.

Every event, traced end to end

Confluent Stream Lineage maps producers, topics, Flink statements and consumers — the live shape of the whole pipeline.

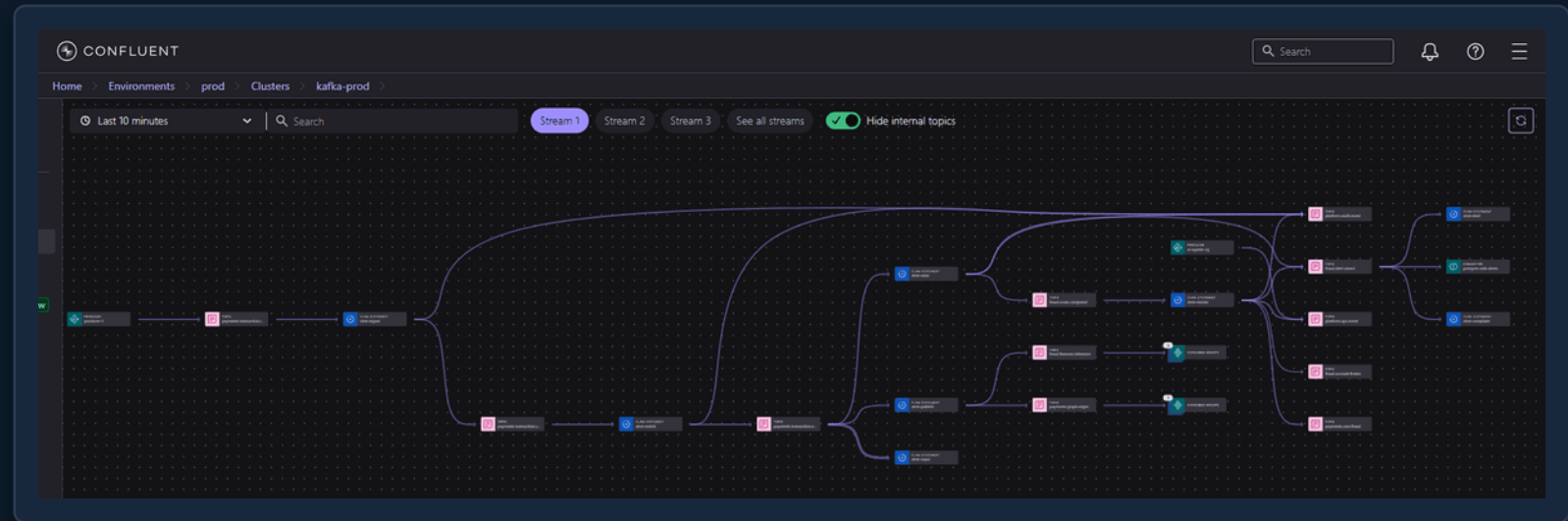
STREAM LINEAGE

The whole pipeline, mapped end to end

• LIVE

WHAT YOU'RE SEEING

- Producers → topics → Flink → consumers
- Every hop visualised and clickable
- Real-time and point-in-time views
- Auto-built from the running streams



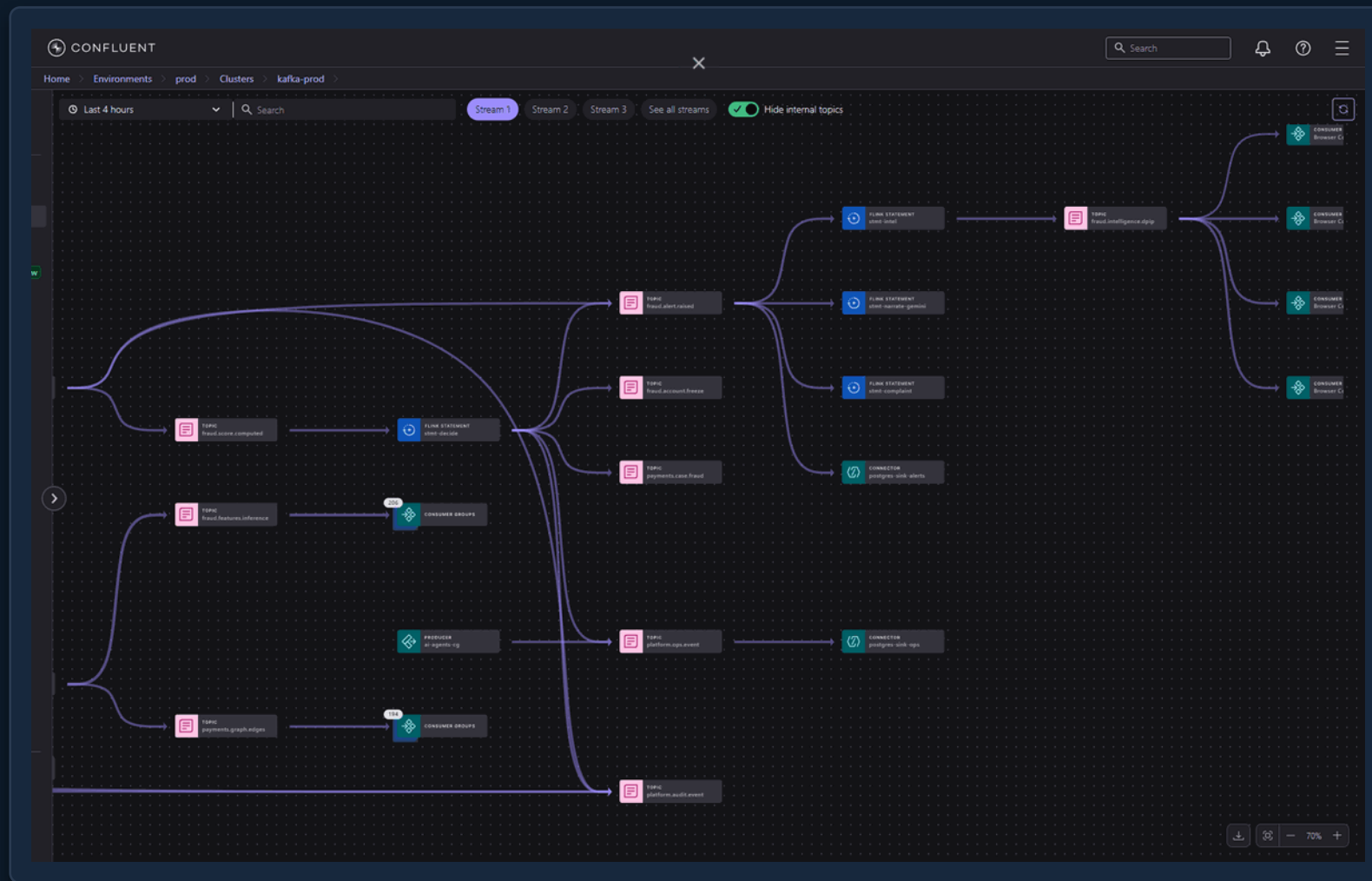
WHY IT MATTERS *You can see — and audit — exactly how every event flows.*

Every decision fans out from STMT-DECIDE

WHAT YOU'RE SEEING

- One statement emits audit, ops & alert events
- fraud.alert.raised feeds sink, intel & complaint
- Atomic, consistent multi-topic output
- Backed by a single Flink checkpoint

WHY IT MATTERS *One decision produces many consistent outputs — never a partial state.*



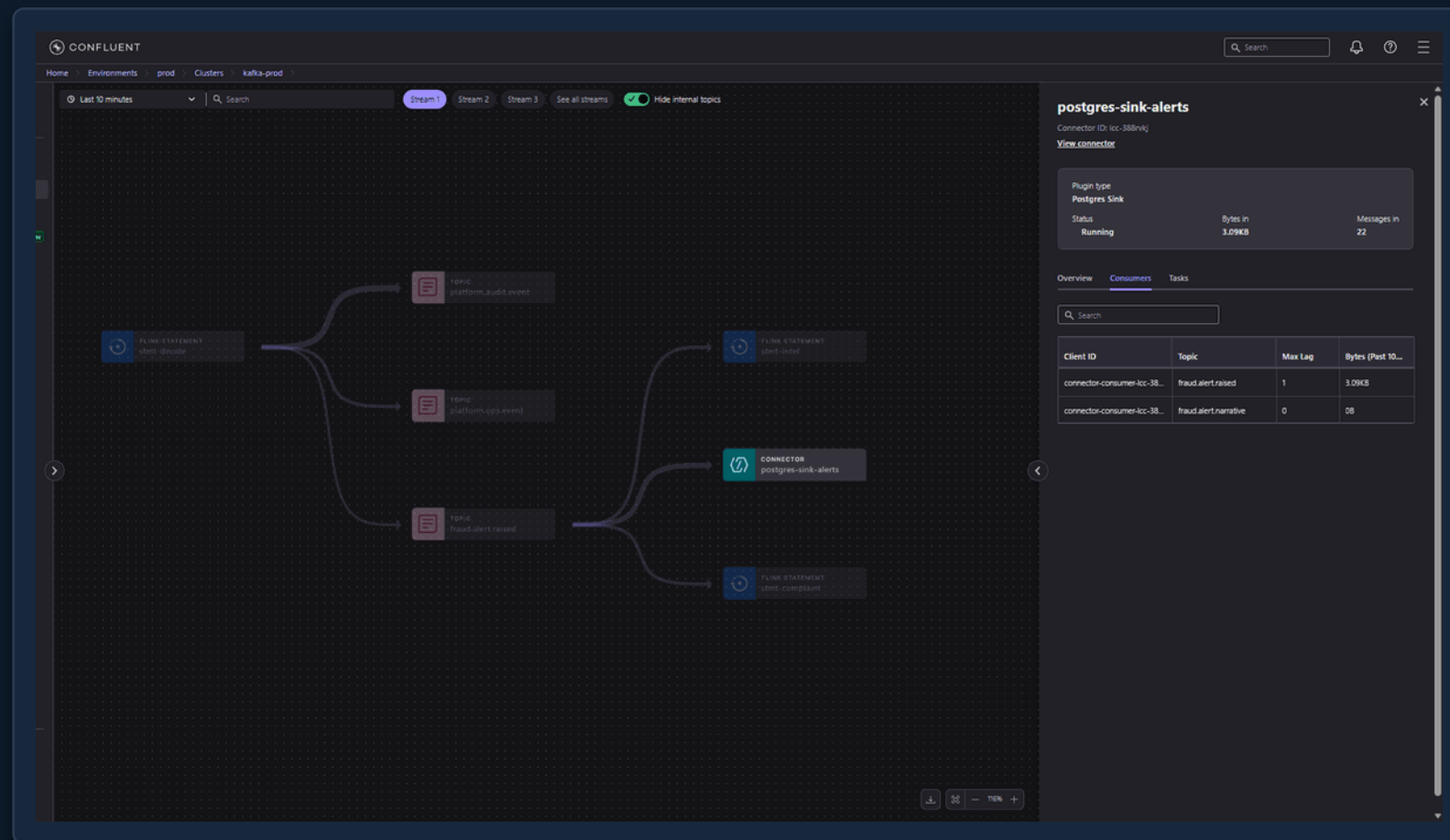
Alerts flow to the operational store

• LIVE

WHAT YOU'RE SEEING

- fraud.alert.raised → postgres-sink-alerts
- Connector health and status shown inline
- Drives the live Fraud Ops dashboard
- Lineage proves the end-to-end path

WHY IT MATTERS Operational egress is observable and fully managed.



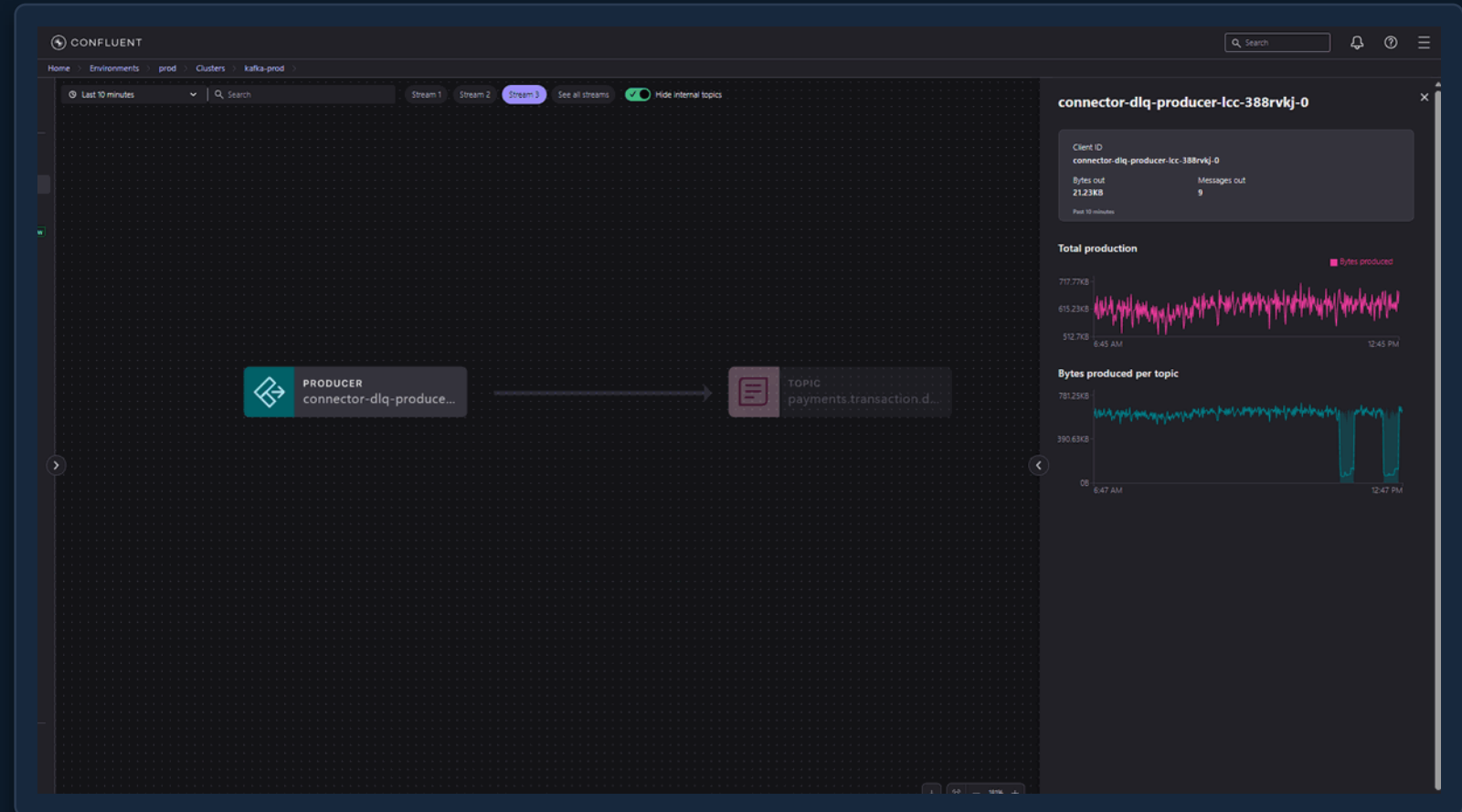
Bad events are quarantined, not dropped

• LIVE

WHAT YOU'RE SEEING

- A dead-letter producer captures failures
- Routed to dedicated dead-letter topics
- Live byte & message metrics confirm flow
- `errors.tolerance = all` keeps it running

WHY IT MATTERS *Malformed input is isolated for review instead of silently lost.*



Governed contracts for every stream

Advanced Stream Governance and the Schema Registry — every topic is a versioned, compatibility-checked Avro contract.

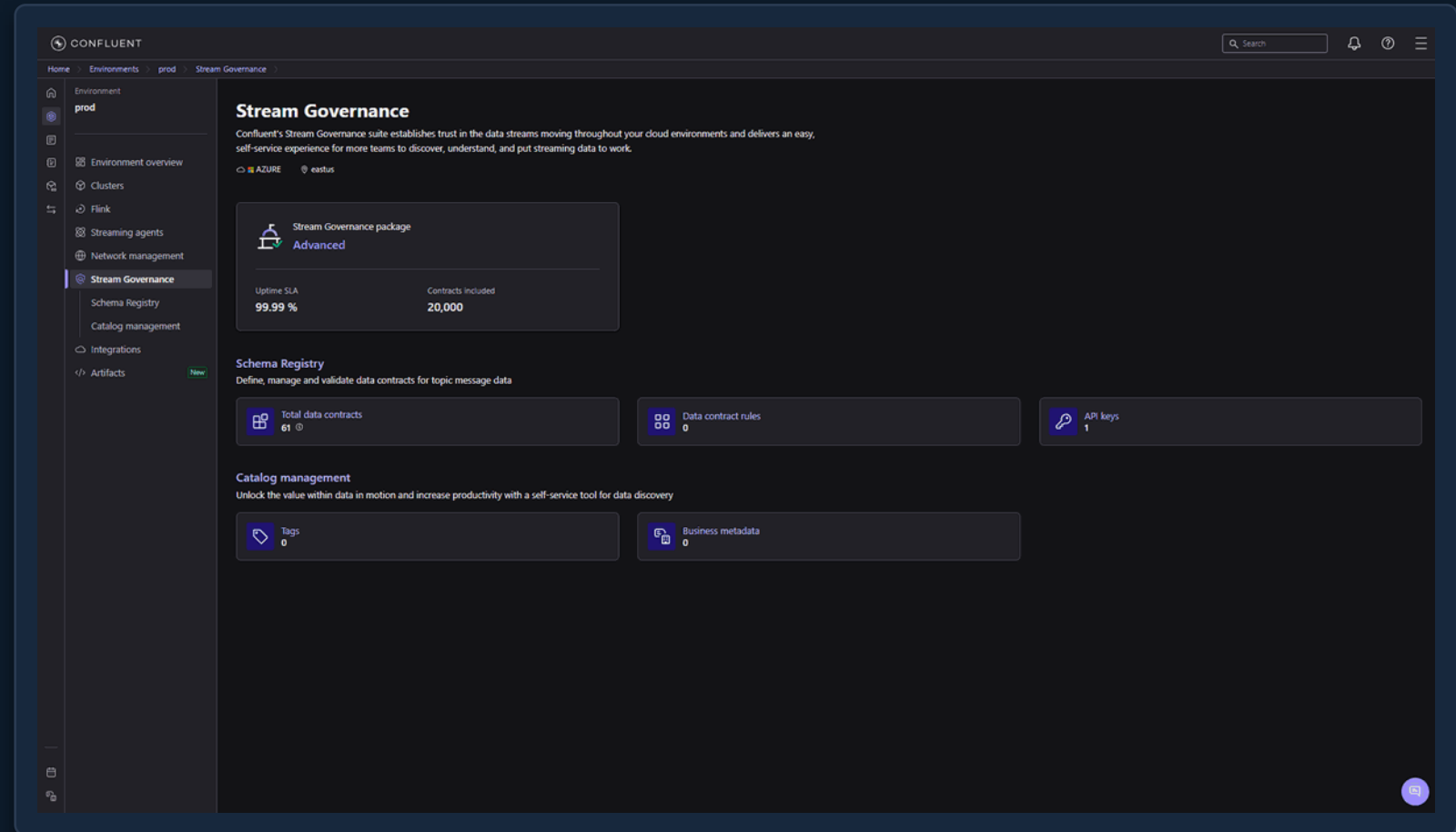
Governance is built into the platform

• LIVE

WHAT YOU'RE SEEING

- Stream Governance package: Advanced
- Schema Registry + Catalog + Lineage
- Data-contract rules enforced
- 99.99% SLA, 20,000 contracts included

WHY IT MATTERS *Trust and discoverability ship with the streams — not bolted on later.*



Every topic is an Avro data contract

WHAT YOU'RE SEEING

- Subjects registered for keys and values
- BACKWARD compatibility enforced
- Lookup, alert & intelligence schemas
- Full version history per subject

WHY IT MATTERS *No producer can publish a change that silently breaks a consumer.*

The screenshot displays the Confluent Schema Registry interface. The left sidebar shows the navigation menu with 'Schema Registry' selected. The main panel is titled 'Schema Registry' and shows the 'Data contracts' tab. It includes a search bar, a compatibility mode dropdown set to 'BACKWARD', and a toggle for 'Show me the latest soft-deleted version'. Below this is a table of registered data contracts.

Subject	Latest version	Schema type	Associated with topics
compliance.sanctions.list-key	1	Avro	View topics
compliance.sanctions.list-value	2	Avro	View topics
customers.account.profile-key	1	Avro	View topics
customers.account.profile-value	2	Avro	View topics
customers.device.profile-key	1	Avro	View topics
customers.device.profile-value	2	Avro	View topics
customers.geo.profile-key	1	Avro	View topics
customers.geo.profile-value	2	Avro	View topics
customers.vulnerability.profile-key	1	Avro	View topics
customers.vulnerability.profile-value	2	Avro	View topics
fraud.account.freeze-value	1	Avro	View topics
fraud.alert.narrative-value	1	Avro	View topics
fraud.alert.raised-value	1	Avro	View topics

The enriched event carries full context

WHAT YOU'RE SEEING

- payments.transaction.enriched-value, Avro
- Validated fields + 11 lookup joins
- Self-contained — no fragile references
- Versioned and Backward-compatible

WHY IT MATTERS Consumers get the full context without re-querying a database.

The screenshot displays the Confluent Schema Registry interface. The left sidebar shows the navigation menu with 'Schema Registry' selected. The main panel shows the details for the schema 'payments.transaction.enriched-value'. The schema is in Avro format, version 1 (current), and is backward-compatible. The schema definition is shown in a code editor, listing fields such as 'event_id', 'txn_id', 'account_id_hash', 'event_time', 'schema_version', 'event_type', and 'beneficiary_add'.

```
1 {
2   "doc": "Output of STMT-ENRICH. Validated fields plus 11-store enrichment context. Not fan-out to STMT-RULES, STMT-PUBLISH, RuleHunter, and AML. No Avro schema references - registers index"
3   "fields": [
4     {
5       "name": "event_id",
6       "type": "string"
7     },
8     {
9       "name": "txn_id",
10      "type": "string"
11    },
12    {
13      "name": "account_id_hash",
14      "type": "string"
15    },
16    {
17      "name": "event_time",
18      "type": {
19        "logicalType": "timestamp-millis",
20        "type": "long"
21      }
22    },
23    {
24      "name": "schema_version",
25      "type": "string"
26    },
27    {
28      "name": "event_type",
29      "type": {
30        "name": "EnrichedEventType",
31        "symbols": [
32          "PAYMENT",
33          "AUTH",
34          "BENEFICIARY_ADD"
35        ],
36        "type": "enum"
37      }
38    }
39  ]
40 }
```

32 schema-governed topics, one per stage

WHAT YOU'RE SEEING

- Lookups, scores, alerts, intelligence, audit
- Each carries a registered data contract
- Network-protected — no ad-hoc edits
- One-click Tableflow to the Iceberg lakehouse

WHY IT MATTERS Every stage of the pipeline is an explicit, governed Kafka stream.

Topic name	Tags	Partitions	Production	Consumption	Retained bytes	Consumers	Data contract	Tableflow sync st
compliance.sanctions.list	--	6	--	--	1.64KB	--	View data contract	Enable Tableflow
customers.account.profile	--	6	--	--	2.04KB	--	View data contract	Enable Tableflow
customers.device.profile	--	6	--	--	2.18KB	--	View data contract	Enable Tableflow
customers.geo.profile	--	6	--	--	2.02KB	--	View data contract	Enable Tableflow
customers.vulnerability.profile	--	6	--	--	1.77KB	--	View data contract	Enable Tableflow
dlq-icc-v7787dn	--	1	5.08KB/s	--	80.52MB	--	Add data contract	Enable Tableflow
dlq-icc-xq96qdx	--	1	4.79KB/s	--	1012.09MB	--	Add data contract	Enable Tableflow
fraud.account.freeze	--	6	7B/s	--	12.86MB	--	View data contract	Enable Tableflow
fraud.alert.narrative	--	6	--	0B/s	0B	--	View data contract	Enable Tableflow
fraud.alert.raised	--	12	606B/s	103.95KB/s	1.19GB	--	View data contract	Enable Tableflow
fraud.alert.state	--	6	--	--	1.72KB	--	View data contract	Enable Tableflow
fraud.features.inference	--	12	303B/s	4.5KB/s	2.14GB	--	View data contract	● Syncing
fraud.intelligence.dpip	--	6	--	--	21.1KB	--	View data contract	● Syncing
fraud.intelligence.molehunter	--	6	--	--	0B	--	View data contract	● Syncing
fraud.score.behavioral	--	12	--	--	0B	--	View data contract	Enable Tableflow
fraud.score.committed	--	12	176B/s	713B/s	667.38KB	--	View data contract	Enable Tableflow

The entry point: every payment event

LIVE

WHAT YOU'RE SEEING

- payments.transaction.raw, on Azure
- PII pre-hashed at the producer
- Lineage: producer → topic → consumers
- Schema panel shows the bound contract

WHY IT MATTERS Raw events enter the platform already PII-safe and schema-validated.

The screenshot shows the Confluent web interface for the 'payments.transaction.raw' topic. The left sidebar contains navigation links for Cluster, Topics, Tableflow, Connectors, Clients, and Kafka Streams. The main content area shows the topic details, including a lineage diagram, a 'Bring your topics to Data Lake' section, and a 'Schema' panel.

Lineage Diagram: The diagram shows the flow of data from the 'producer producer-1' to the 'topic payments.transaction.raw', which then branches out to four consumers: 'mulehunter-rph-cg', 'platform.audit.event', 'smoke-2793498029...', and 'lcc-388rvj'.

Bring your topics to Data Lake: This section provides information on how to expose the topic as an Apache Iceberg table to easily populate Data Lakes. It includes a link to 'Find out here' and an 'Enable Tableflow' button.

Schema: The schema panel shows the 'PaymentTransaction' schema with the following details:

- doc: payments.transaction.raw-v1.0
- namespace: com.infobys.fraud.events
- fields (4)

Managed ingest, egress & lakehouse

Postgres CDC source, Postgres sinks and Tableflow → Apache Iceberg — all fully managed, zero bespoke code.

Managed connectors bridge Kafka ↔ Postgres

WHAT YOU'RE SEEING

- postgres-cdc-source-lookups (source)
- postgres-sink-alerts (sink)
- postgres-sink-ops (sink)
- All fully managed, all Running

WHY IT MATTERS *Ingest and egress are managed connectors — zero bespoke code.*

The screenshot displays the Confluent Connectors management interface. The left sidebar shows the navigation menu with 'Connectors' selected. The main panel, titled 'Connectors', lists three active connectors:

Connector Name	Status	Tasks	Bytes/sec	Messages/sec	Messages behind
postgres-cdc-source-lookups	Running	1	0B/s	0	--
postgres-sink-alerts	Running	6	16.28KB/s	42	4,309,658
postgres-sink-ops	Running	1	0B/s	0	--

Below the list, there is a section 'Connect with popular connectors' featuring tiles for various connectors like Snowflake Sink, Azure Blob Storage Sink, Elasticsearch Service Sink, MongoDB Atlas Sink, Amazon Kinesis Source, Azure Data Lake Storage Gen2 Sink, MongoDB Atlas Source, and Salesforce CDC Source.

CDC streams lookup data into Kafka

WHAT YOU'RE SEEING

- PostgresCdcSourceV2, status Running
- 10 compacted lookup topics
- Feeds FOR SYSTEM_TIME AS OF joins
- Health & throughput metrics inline

WHY IT MATTERS *Reference data lands in Kafka in real time for enrichment.*

The screenshot shows the Confluent UI for a connector named 'postgres-cdc-source-lookups'. The connector is in a 'Running' state. The left sidebar shows the navigation menu with 'Connectors' selected. The main panel displays the connector's status, health metrics, and performance metrics.

Connector health

Running	Unassigned	User failure	System failure	Stopped
1	0	0	0	0

Performance metrics (last 7 days)

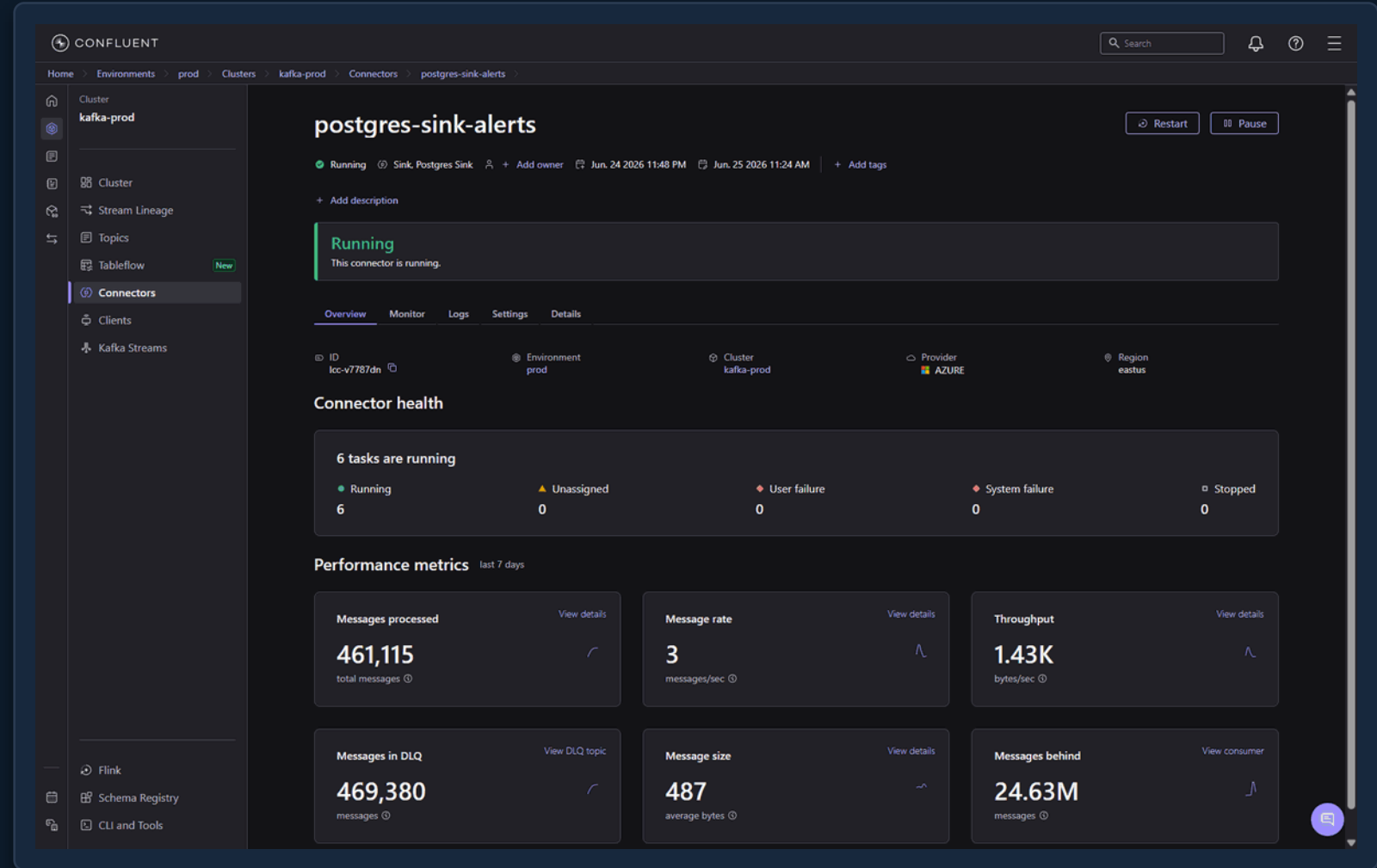
Metric	Value	Unit
Messages processed	99	total messages
Message rate	0	messages/sec
Throughput	1	bytes/sec
Message size	0	average bytes

Fraud decisions land in PostgreSQL

WHAT YOU'RE SEEING

- postgres-sink-alerts, status Running
- Consumes fraud.alert.raised + narrative
- UPSERT keyed by alert_id
- Powers the dashboard & case management

WHY IT MATTERS *The operational store is kept live by a managed sink.*



Zero-ETL streams to an open lakehouse

WHAT YOU'RE SEEING

- fraud.features.inference → Apache Iceberg
- One declarative config, not 4 connectors
- Tableflow materialises analytical topics
- Open table format for any engine

WHY IT MATTERS *Analytical exhaust reaches the lakehouse with zero ETL.*

The screenshot displays the Confluent Kafka management console for the 'kafka-prod' cluster. The main view is for the topic 'fraud.features.inference'. The left sidebar shows the navigation menu with 'Topics' selected. The main content area shows the topic's configuration, including 'Tableflow syncing' status, 'Iceberg' format, and various tabs like Overview, Messages, Monitor, Data contract, Access, Settings, and Details. The 'Overview' tab is active, showing production and consumption rates. Below this, a 'Stream Lineage' diagram illustrates the data flow from a producer to a Tableflow connector and then to a consumer. The 'Tableflow details' section lists configuration parameters such as Endpoint, Namespace, Table name, Catalog, Provider, and Encryption. The 'Schema' section shows the schema for 'FraudFeaturesInference'.

fraud.features.inference

Tableflow syncing Data contract associated Add owner Jun. 24 2026 12:35 PM Add tags

+ Add description

Overview Messages Monitor Data contract Access Settings Details

Environment: prod Cluster: kafka-prod Provider: AZURE Region: eastus Production: 1.73KB/s Consumption: 0B/s Partitions: 12

Period: Last hour Refresh

Stream Lineage: producer-producer-1 → tableflow-fraud-features-inference → consumer-consumer-null-4015

Tableflow details

Endpoint	Not available
Namespace	lkc-7yy59x1
Table name	fraud.features.inference
Catalog	Not available
Provider	Confluent
Encryption	Automatic

Schema [View full schemas](#)

FraudFeaturesInference (0)

- doc:** Hashed feature vector for the external AI inference service. All identifiers HMAC-hashed at source — no raw PII.
- namespace:** com.infosys.fraud.events
- fields (0)**

Nine SQL statements, one decision core

The deterministic detection engine — ingest to decide — plus governed AI narration, all in Confluent Flink.

Nine SQL statements are the detection core

WHAT YOU'RE SEEING

- ingest → enrich → rules → cepxr → decide → narrate
- plus stmt-publish, stmt-intel and stmt-complaint
- all Running · ~1 CFU each · zero lag
- deployed as EXECUTE STATEMENT SET jobs

WHY IT MATTERS *The whole fraud engine is declarative, observable Flink SQL.*

The screenshot shows the Confluent Flink management console. On the left is a sidebar with navigation options like Environment, Clusters, and Flink. The main panel displays a table of Flink statements. The table has 7 columns: Flink statement name, Status, Statement type, Statement CFU, State size GB, Created, and # Messages behind total. There are 10 statements listed, all of which are 'EXECUTE STATEMENT SET' jobs. Most are in a 'Running' state, but 'stmt-cepxr' is in a 'Degraded' state. All other statements show 0 messages behind.

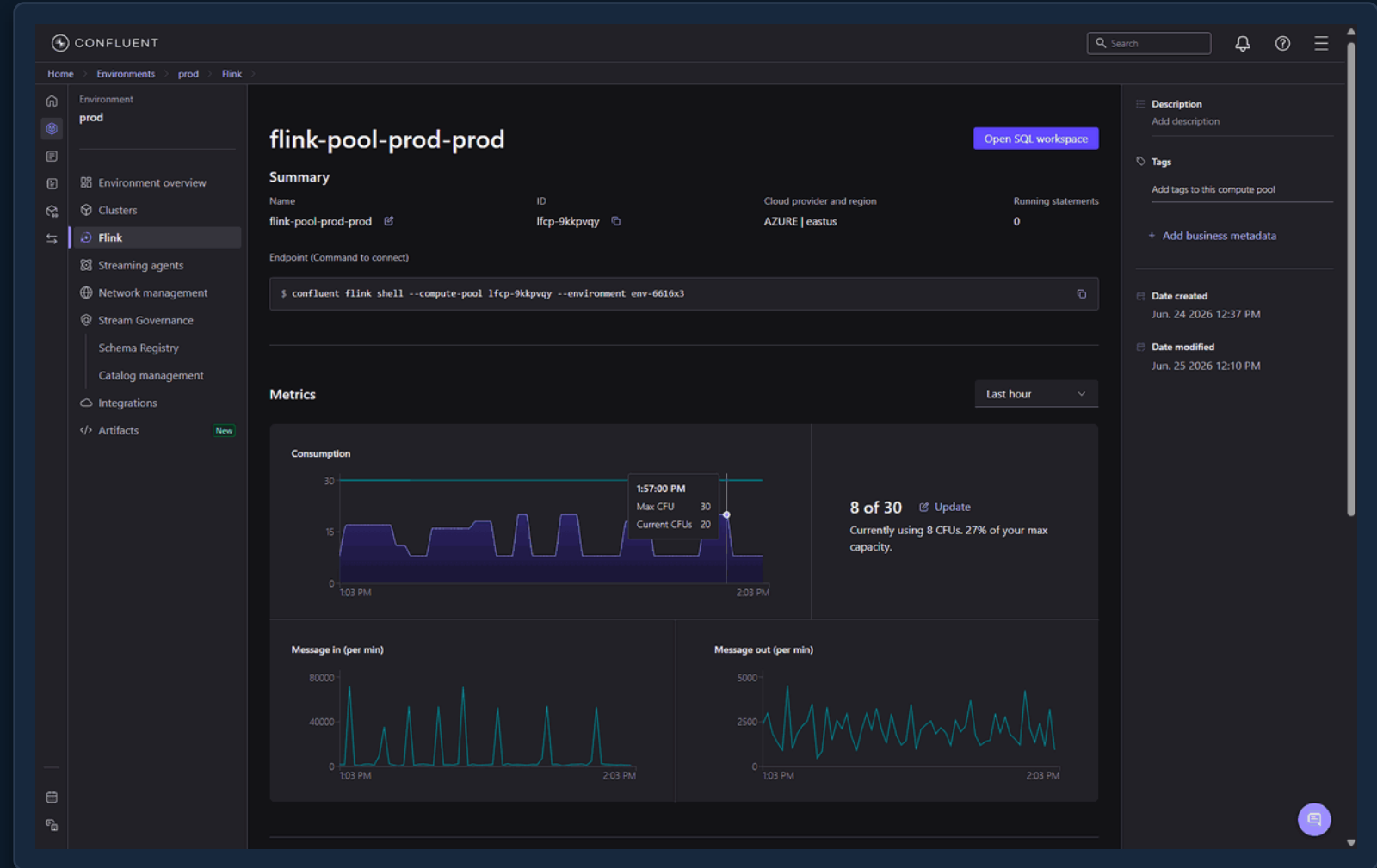
Flink statement name	Status	Statement type	Statement CFU	State size GB	Created	# Messages behind total
stmt-enrich flink-pool-prod-prod	Running	EXECUTE STATEMENT SET	1	0.01	14 hours ago	0 Not behind
stmt-publish flink-pool-prod-prod	Running	EXECUTE STATEMENT SET	1	-	a day ago	0 Not behind
stmt-ingest flink-pool-prod-prod	Running	EXECUTE STATEMENT SET	1	1.34	a day ago	0 Not behind
stmt-rules flink-pool-prod-prod	Running	EXECUTE STATEMENT SET	1	-	a day ago	0 Not behind
stmt-decide flink-pool-prod-prod	Running	EXECUTE STATEMENT SET	1	0	a day ago	0 Not behind
stmt-complaint flink-pool-prod-prod	Running	EXECUTE STATEMENT SET	1	0.85	a day ago	0 Not behind
stmt-cepxr flink-pool-prod-prod	Degraded	EXECUTE STATEMENT SET	1	-	a day ago	-
stmt-intel flink-pool-prod-prod	Running	EXECUTE STATEMENT SET	1	0	a day ago	0 Not behind

Elastic compute that scales with load

WHAT YOU'RE SEEING

- flink-pool-prod-prod on Azure eastus
- Using 8 of 30 CFUs — ~27% of capacity
- Live consumption & messages metrics
- Scales up under burst, back down when idle

WHY IT MATTERS *You pay for fraud-processing capacity only as it is actually needed.*



AI narration runs inside Flink — live

WHAT YOU'RE SEEING

- CREATE MODEL gemini_narrate (Google Gemini)
- ML_PREDICT invoked straight from Flink SQL
- A HOLD alert → a plain-English note in seconds
- PII-free and stamped with the model id

WHY IT MATTERS *Post-decision AI explains the alert — it never makes the decision.*

The screenshot displays the Confluent Flink SQL IDE interface. The left sidebar shows a workspace tree with folders for 'examples', 'marketplace', 'prod', 'dev', and 'scam-shield'. The main editor area shows two SQL queries and their execution results.

Query 1:

```
1 CREATE MODEL 'gemini_narrate'
2 INPUT (prompt STRING)
3 OUTPUT (response STRING)
4 WITH (
5   'provider' = 'googleai',
6   'task' = 'text_generation',
7   'googleai.connection' = 'gemini-conn',
8   'googleai.system_prompt' =
9     'You are a fraud analyst. In AT MOST 2 short sentences, explain why the
10    described transaction was flagged with the given final_action. Be factual.
11    Do NOT use the customer''s name or any account identifier.'
12 );
```

Execution Result 1:

workspace-2026-06-23-0814... Completed Jun. 23, 2026, 3:43 PM Mode Streaming Run

Success

Model 'gemini_narrate' with version '2' created

Query 2:

```
1 SELECT t.prompt, p.response
2 FROM (VALUES ('Alert A1 | Action: HOLD | Vulnerability tier: DIGITAL_VULNERABLE | Reason code: APP_NEWBENE_DWELL')) AS t(prompt)
3 CROSS JOIN LATERAL TABLE(ML_PREDICT('gemini_narrate', t.prompt)) AS p;
```

Execution Result 2:

workspace-2026-06-23-0814... Completed Jun. 23, 2026, 3:42 PM 1 Mode Streaming Run

prompt	response
Alert A1 Action: HOLD...	The transaction was placed on hold because a digitally vulnerable customer attempted a transfer to a newly added beneficiary via the mobile app. The system flagge...

Last saved at 6/23/26, 3:43 PM 2 / 20 cells

Every service, itemised and metered

WHAT YOU'RE SEEING

- Flink CFUs, Kafka cluster & Connect tasks itemised
- Advanced Stream Governance billed hourly
- Private-link networking and storage broken out
- Per-resource consumption visible for planning

WHY IT MATTERS Full line-item visibility into exactly which services the platform consumes.

The screenshot displays the Confluent Billing and payment interface. The left sidebar contains navigation links: Home, Environments, Data Portal, SQL workspaces, Cluster links, and Migration hub (marked as New). The main content area is titled 'Billing and payment' and shows 'Invoice period details' for 'environment - prod (\$188.98 USD)'. Below this, 'Non-cluster-specific charges' are listed in a table:

Resource name	Quantity	Unit price	Total
flink-pool-prod-prod (llcp-9kpvqy)	FlinkNumCFUs: 249.5500 CPU-hour	\$0.21 / CPU-hour	\$52.41
kafka-prod (llc-7yy59x1)	ConnectCapacity: 13.0000 Hour	\$0.27777778 / Hour	\$3.61
Always On Stream Governance Package (lsr-c33kndj)	GovernanceBase: 25.0000 Hour	\$1 / Hour	\$25.00

Below the table, the 'Kafka cluster ID: llc-7yy59x1' is shown with a detailed table of resource usage:

Resource name	Quantity	Unit price	Total
postgres-cdc-source-lookups (lcc-pggwg2k)	ConnectNumTasks: 13.0000 Task-hour	\$0.53 / Task-hour	\$6.89
postgres-sink-alerts (lcc-v7787dn)	ConnectNumTasks: 23.0000 Task-hour	\$0.13416667 / Task-h...	\$3.09
postgres-sink-ops (lcc-xqq6qdx)	ConnectNumTasks: 18.0000 Task-hour	\$0.13416667 / Task-h...	\$2.42
kafka-prod (llc-7yy59x1)	KafkaStorage: 414.8087 GB-hour	\$0.00010959 / GB-h...	\$0.05
	KafkaNumCKUs: 26.0000 CKU-hour	\$3.2351 / CKU-hour	\$84.11
	KafkaNetworkWrite, private-link: 23.9055 GB	\$0.04 / GB	\$0.96
	KafkaNetworkRead, private-link: 19.7590 GB	\$0.04 / GB	\$0.79
	TableflowDataProcessed: 1.5902 GB	\$0.04 / GB	\$0.06
	TableflowNumTopics: 96.0000 Topic-hour	\$0.10 / Topic-hour	\$9.60

At the bottom left, there are links to 'Schedule a help call' and 'File Support ticket'. A chat icon is visible in the bottom right corner.

Demo-to-production, entirely as code

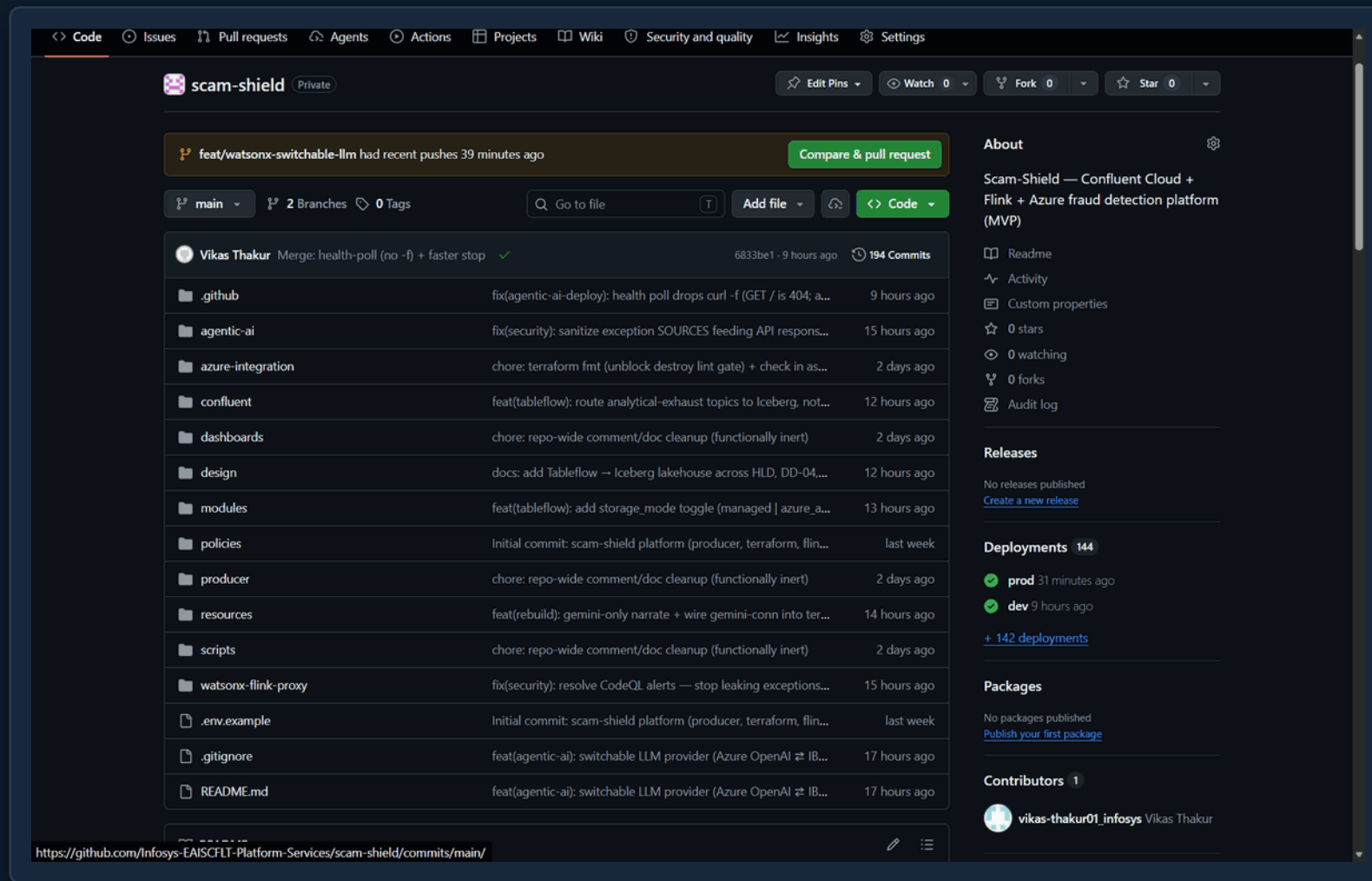
Two Terraform roots in Git, planned, reviewed and applied through GitHub Actions across Confluent and Azure.

Everything is code, in one repo

WHAT YOU'RE SEEING

- confluent/ and azure-integration/ roots
- modules/, resources/, policies/ alongside
- No resource ever created by hand
- Drift reconciled back to Git on every run

WHY IT MATTERS *Terraform is the sole source of truth — if it's not in Git, it doesn't exist.*



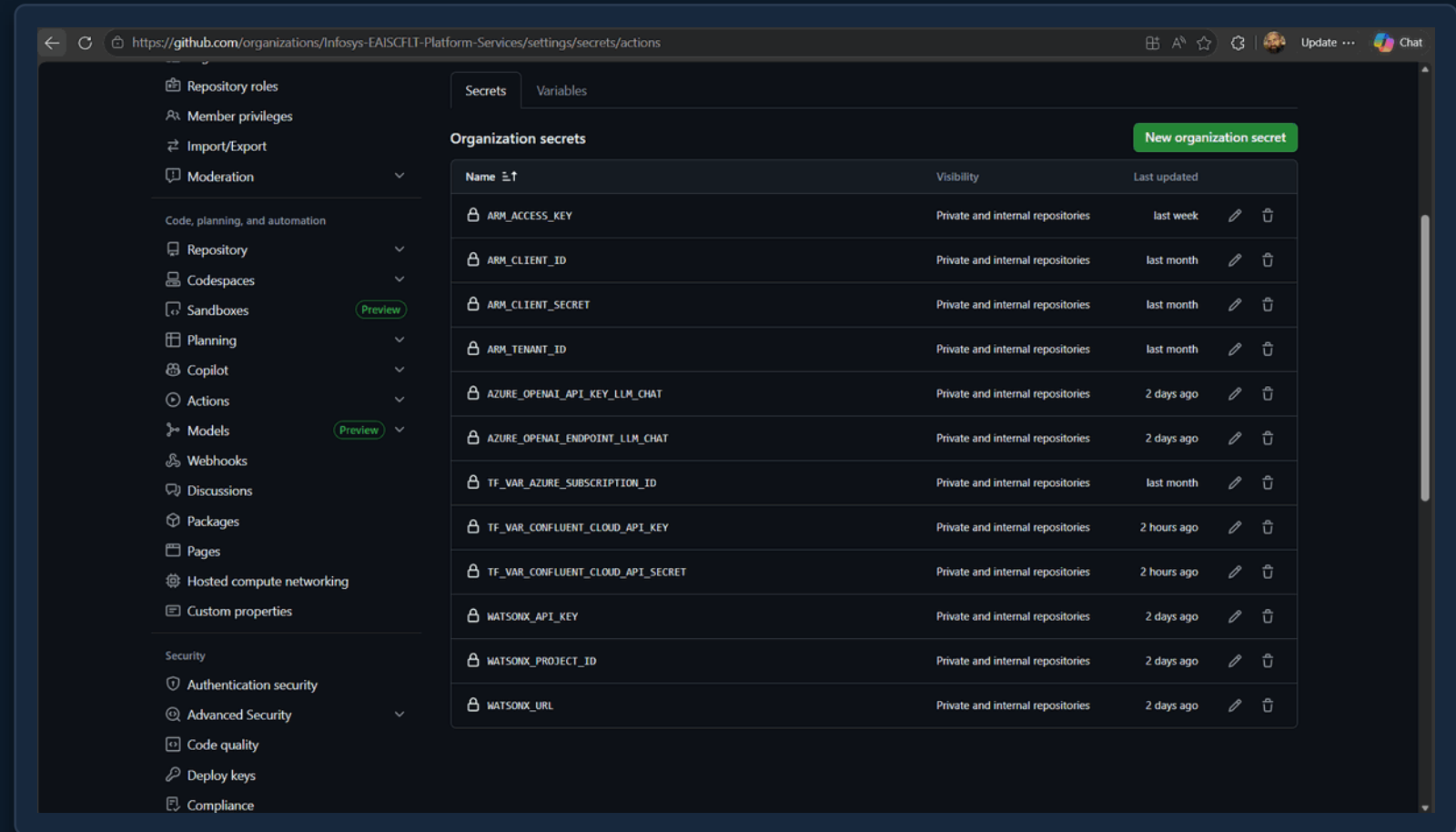
Config is committed; secrets never are

• LIVE

WHAT YOU'RE SEEING

- tfvars and the lock file are committed
- Secrets live only in GitHub Secrets
- Injected via `TF_VAR_*` at run time
- Marked `sensitive = true`

WHY IT MATTERS *A clean split of committed config from injected secrets.*



Lint → plan → approve → apply

WHAT YOU'RE SEEING

- Every change flows through GitHub Actions
- Format-check & validate gate each commit
- Plans reviewed before any apply
- prod / prod-dr applies require approval

WHY IT MATTERS *No one applies from a laptop — promotion is a reviewed, auditable pipeline.*

The screenshot displays the GitHub Enterprise Actions interface for the repository 'Infosys-EAISCFLT-Platform-Services / scam-shield'. The 'Actions' tab is selected, showing a list of workflow runs. A notification banner at the top indicates an upcoming change to the GitHub App installation token format. The 'All workflows' section shows 573 workflow runs. The table below lists the most recent runs:

Workflow	Event	Status	Branch	Actor	Run Time
Terraform — Confluent Cloud	Manually run by vikas-thakur01_infosys	Success	feat/watsonx-switchable-11m	vikas-thakur01_infosys	40 minutes ago, 12m 51s
Terraform — Confluent Cloud	Manually run by vikas-thakur01_infosys	Success	feat/watsonx-switchable-11m	vikas-thakur01_infosys	42 minutes ago, 50s
Terraform — Confluent Cloud	Manually run by vikas-thakur01_infosys	Success	feat/watsonx-switchable-11m	vikas-thakur01_infosys	47 minutes ago, 50s
Terraform — Confluent Cloud	Manually run by vikas-thakur01_infosys	Success	main	vikas-thakur01_infosys	Today at 8:54 AM, 1m 18s
Terraform — Confluent Cloud	Manually run by vikas-thakur01_infosys	Success	main	vikas-thakur01_infosys	Today at 8:52 AM, 46s
Terraform — Confluent Cloud	Manually run by vikas-thakur01_infosys	Success	main	vikas-thakur01_infosys	Today at 8:32 AM, 54s
Agentic AI — Build & Deploy to Jumphost	Manually run by vikas-thakur01_infosys	Success	main	vikas-thakur01_infosys	Today at 12:17 AM, 2m 34s

Every change is a reviewed plan first

WHAT YOU'RE SEEING

- terraform plan runs per environment
- add / change / destroy shown explicitly
- The saved plan is exactly what applies
- Daily scheduled drift detection on prod

WHY IT MATTERS *Changes are previewed and approved before they touch the platform.*

The screenshot displays the Terraform Plan interface. On the left, a sidebar shows a list of jobs under 'All jobs'. The 'Plan — prod' job is highlighted. The main panel shows the details of the 'Plan — prod' job, which succeeded 1 hour ago in 21s. The job output is a Terraform plan for the 'prod' environment, showing a list of resources to be added or updated. The resources include:

- `cost_center` = "CC-FRAUD-002"
- `description` = "STMT-RULES -- folds 308-05 (UPI), 308-06 (Card), 308-07 (Transfer) rule engines"
- `domain` = "fraud-detection"
- `sa_id` = (known after apply)
- `team` = "fraud-platform"
- `service_account_ids` = {
 - `agentic-ai-consumer` = (known after apply)
 - `agentic-ai-service` = (known after apply)
 - `ai-inference-service` = (known after apply)
 - `aml-engine` = (known after apply)
 - `aml-investigations-consumer` = (known after apply)
 - `app-manager` = (known after apply)
 - `audit-blob-sink-connector` = (known after apply)
 - `case-management-consumer` = (known after apply)
 - `cbs-freeze-consumer` = (known after apply)
 - `complaint-intake-api` = (known after apply)
 - `compliance-consumer` = (known after apply)
 - `fraud-producer` = (known after apply)
 - `graphdb-sink-connector` = (known after apply)
 - `graphdb-source-connector` = (known after apply)
 - `lookup-loader` = (known after apply)
 - `mulehunter-scoring-service` = (known after apply)
 - `mulehunter-training-export-connector` = (known after apply)
 - `notification-service` = (known after apply)
 - `npci-dpip-export-connector` = (known after apply)
 - `ops-dashboard-consumer` = (known after apply)
 - `ops-remediation-consumer` = (known after apply)
 - `sa-conn-pg-cdc-lookups` = (known after apply)
 - `sa-conn-pg-sink-alerts` = (known after apply)
 - `sa-flink-stmt-cepxr` = (known after apply)
 - `sa-flink-stmt-complaint` = (known after apply)
 - `sa-flink-stmt-decide` = (known after apply)
 - `sa-flink-stmt-enrich` = (known after apply)
 - `sa-flink-stmt-ingest` = (known after apply)
 - `sa-flink-stmt-intel` = (known after apply)
 - `sa-flink-stmt-narrate` = (known after apply)
 - `sa-flink-stmt-publish` = (known after apply)
 - `sa-flink-stmt-rules` = (known after apply)

Schemas, topics, Flink & connectors — applied

WHAT YOU'RE SEEING

- confluent/ apply provisions the whole estate
- cluster, topics, schemas, Flink, connectors
- Consumes the approved plan automatically
- Outputs feed the Azure integration root

WHY IT MATTERS *The entire Confluent platform stands up from code.*

The screenshot displays the Terraform web interface. On the left, a sidebar titled 'Summary' shows a list of jobs. The 'Apply — prod' job is selected and highlighted. Below the job list, there are links for 'Run details', 'Usage', and 'Workflow file'. The main panel on the right shows the detailed output of the 'Apply — prod' job, which started 17m 30s ago. The output is a log of Terraform actions, including setting up the job, running actions/checkout, running hashicorp/setup-terraform, and the main 'Terraform Apply' step. The 'Terraform Apply' step is expanded, showing a list of resources being created or updated, such as 'module.security.confluent_role_binding.this["agentic-ai-service-0"]', 'module.security.confluent_role_binding.this["sa-flink-stmt-enrich-1"]', and 'module.security.confluent_role_binding.this["lookup-loader-2"]'. Each resource is shown with its status (e.g., 'Still creating...') and the time elapsed (e.g., '[10s elapsed]').

Azure applied from the same pipeline

WHAT YOU'RE SEEING

- azure-integration/ reads Confluent state
- Key Vault, Private Link, Postgres, monitoring
- Apply with -auto-approve in CI
- Outputs hand off cleanly between roots

WHY IT MATTERS Both clouds provisioned in one reviewed pipeline.

The screenshot displays a GitHub Actions workflow run for the 'Apply Azure — prod' job. The job is marked as successful and completed 11m 47s ago. The logs show the execution of Terraform commands to apply Azure resources.

Summary

- Resolve Environments
- Lint & Validate — confluent
- Lint & Validate — azure-integration
- Lint & Validate — aws-integration
- Plan — \${{ matrix.environment }}
- Destroy — \${{ matrix.environment }}
- Drift Detection — prod
- Plan Azure — prod
- Seed Postgres DDL — \${{ matrix.environment }}
- Destroy Azure — \${{ matrix.environment }}
- Plan AWS — \${{ matrix.environment }}
- Destroy AWS — \${{ matrix.environment }}
- Apply — \${{ matrix.environment }}
- Apply Azure — prod**
- Apply AWS — \${{ matrix.environment }}

Run details

- Usage
- Workflow file

Apply Azure — prod
succeeded 25 minutes ago in 11m 59s

Terraform Apply — azure-integration 11m 47s

```

127 azurerm_private_dns_a_record.ccn_wildcard[0]: Creation complete after 1s [id=/subscriptions/***/resourceGroups/rg-confluent-prod/providers/Microsoft.Network/privateDnsZones/g0do07.eastus.azure.confluent.cloud/A/*]
128
129 | Warning: Applied changes may be incomplete
130 |
131 | The plan was created with the -target option in effect, so some changes
132 | requested in the configuration may have been ignored and the output values
133 | may not be fully updated. Run the following command to verify that no other
134 | changes are pending:
135 |   terraform plan
136 |
137 | Note that the -target option is not suitable for routine use, and is
138 | provided only for exceptional situations such as recovering from errors or
139 | mistakes, or when Terraform specifically suggests to use it as part of an
140 | error message.
141 |
142
143 Apply complete! Resources: 7 added, 0 changed, 5 destroyed.
144
145 Outputs:
146
147 action_group_id = "/subscriptions/***/resourceGroups/rg-confluent-prod/providers/Microsoft.Insights/actionGroups/ag-confluent-prod"
148 ccn_private_endpoint_ip = "100.78.217.29"
149 confluent_network_dns_domain = "g0do07.eastus.azure.confluent.cloud"
150 confluent_network_id = "n-g0do07"
151 hackathon_postgres_admin_password = <sensitive>
152 hackathon_postgres_admin_username = "scamadmin"
153 hackathon_postgres_database_name = "scamshield"
154 hackathon_postgres_fqdn = "psql-scamshield-prod-ppc.postgres.database.azure.com"
155 hackathon_postgres_server_id = "/subscriptions/***/resourceGroups/rg-confluent-prod/providers/Microsoft.DBforPostgreSQL/flexibleServers/psql-scamshield-prod-ppc"
156 log_analytics_workspace_id = "/subscriptions/***/resourceGroups/rg-confluent-prod/providers/Microsoft.OperationalInsights/workspaces/law-confluent-prod"
157 log_analytics_workspace_name = "law-confluent-prod"
158 resource_group_id = "/subscriptions/***/resourceGroups/rg-confluent-prod"
159 workbook_id = "/subscriptions/***/resourceGroups/rg-confluent-prod/providers/Microsoft.Insights/workbooks/b4bb9e69-98d1-5aa1-9b74-73afc3ca6bc"
  
```

Post Run actions/checkout@v4 0s

Complete job 0s

<https://github.com/Infosys-EAISCFLT-Platform-Services/scam-shield/actions/runs/28073978849/job/83114296837>

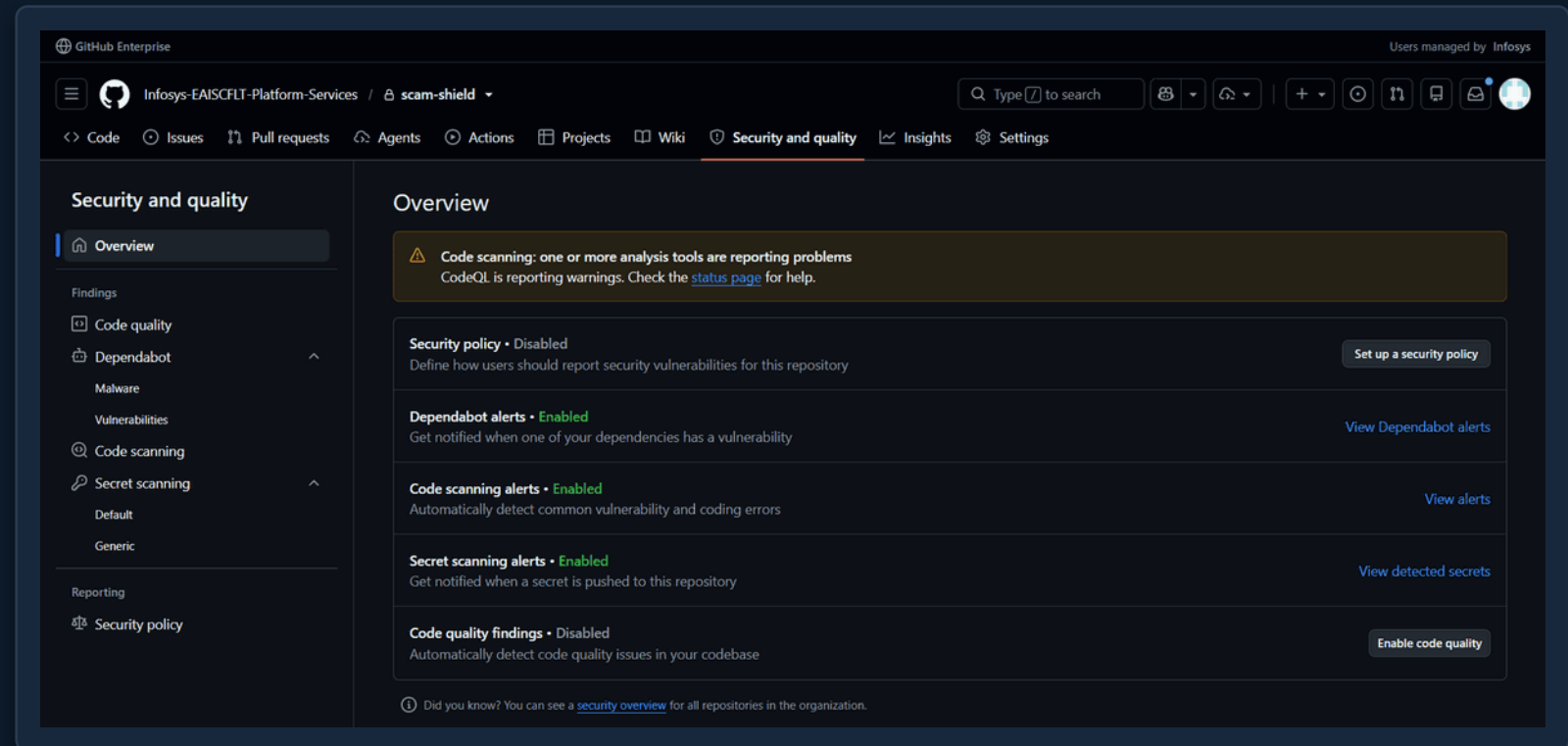
Clean code, enforced automatically

• LIVE

WHAT YOU'RE SEEING

- Secret scanning blocks credentials
- CodeQL flags vulnerabilities on push
- Dependabot tracks dependency risk
- OPA / Conftest gates non-compliant plans

WHY IT MATTERS *Security and policy are enforced in CI, not left to a reviewer's memory.*



What the analyst actually sees

The governed agentic layer on top — a live command center that triages, investigates, audits and learns under HITL.

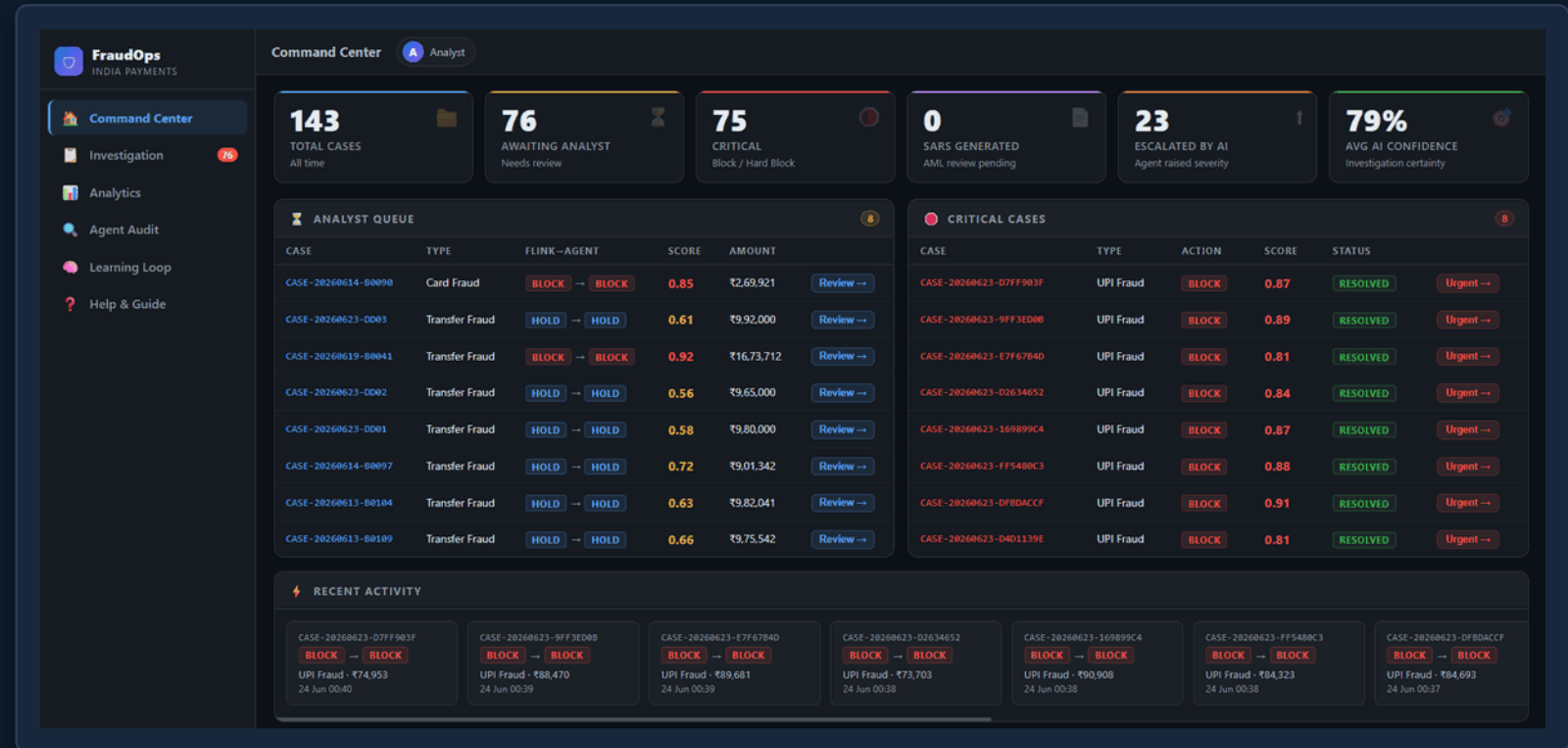
A live command center for fraud ops

• LIVE

WHAT YOU'RE SEEING

- Tiles: total, awaiting analyst, critical, escalated
- 79% auto-handled; the rest queued for a human
- Analyst queue ranked by risk, action & SLA
- Fed by fraud.alert.raised + ops events over SSE

WHY IT MATTERS Analysts see the whole fraud picture and act where it matters most.



Deterministic score, agent cross-check

WHAT YOU'RE SEEING

- Flink score & agent confidence side by side
- Decision severity path ALLOW → HARD_BLOCK
- Fraud patterns & per-rule validation listed
- The agent corroborates — it never re-scores

WHY IT MATTERS *The agent assists the deterministic decision; it never replaces it.*

FraudOps
INDIA PAYMENTS

Investigation Workspace 143

Search case ID, txn, alert...

Block Hold Challenge Escalated
SAR Awaiting

Risk Score Meters

FLINK SCORE 87%

AGENT CONFIDENCE 92%

RISK LEVEL CRITICAL

Decision Severity Path

ALLOW CHALLENGE HOLD BLOCK F+A HARD BLOCK

■ Flink decision ■ Agent recommendation ■ Both agreed

Fraud Patterns Detected

SIM SWAP TAKEOVER

Rule Validation

SIM-SWAP-001
UNCONFIRMED N/A
No additional detail

APP-002
UNCONFIRMED N/A
No additional detail

APP-001-TFR
UNCONFIRMED N/A
No additional detail

MULE-001
UNCONFIRMED N/A
No additional detail

VEL-001
✓ CONFIRMED CRITICAL
No additional detail

Investigation Narrative

Investigation Findings

Analyst Narrative for UPI Transaction TXN-UPI-20260623-P0010

1. Rules Fired and Their Significance - VEL-001 (CRITICAL): This rule fired because the transaction amount (Rs74,953) is 749.53 times the account's 30-day average, exceeding the >100x threshold. This extreme deviation from the account's typical behavior is a strong indicator of potential fraudulent activity, as legitimate transactions rarely deviate so drastically from established patterns. The rule's confirmation and critical severity underscore the need for immediate action to prevent potential financial loss and account compromise.

● Live · 10:23

Every alert gets a plain-English narrative

WHAT YOU'RE SEEING

- A structured investigation narrative per case
- Cites the rules, signals & recommended action
- All model calls via the governance gateway
- PII-free — the AI never sees raw identifiers

WHY IT MATTERS *Investigators read why, not just a score — governed and auditable.*

The screenshot displays the FraudOps India Payments Investigation Workspace. The interface includes a sidebar with navigation options: Command Center, Investigation (76), Analytics, Agent Audit, Learning Loop, and Help & Guide. The main area is titled 'Investigation Workspace' and shows a list of 143 cases. Each case entry includes a status (e.g., BLOCK, CRITICAL), a case ID, a transaction ID, a fraud type (UPI Fraud), and a score (e.g., 0.87).

The right panel provides a detailed investigation narrative for a selected case. It includes sections for 'Investigation Findings' and 'Additional Context'. The 'Investigation Findings' section contains a structured narrative for a UPI Transaction, detailing the rules fired, behavioral signals, and recommended actions. The 'Additional Context' section provides further details about the transaction and the rules triggered.

Investigation Narrative

Investigation Findings

Analyst Narrative for UPI Transaction TXN-UPI-20260623-P0010

- Rules Fired and Their Significance** - VEL-001 (CRITICAL): This rule fired because the transaction amount (Rs74,953) is 749.53 times the account's 30-day average, exceeding the >100x threshold. This extreme deviation from the account's typical behavior is a strong indicator of potential fraudulent activity, as legitimate transactions rarely deviate so drastically from established patterns. The rule's confirmation and critical severity underscore the need for immediate action to prevent potential financial loss and account compromise.
- Behavioral Signals - Supporting Signals** - The high behavioral risk score aligns with the VEL-001 rule's findings, suggesting a potential SIM swap takeover attempt. This pattern is consistent with unauthorized access attempts where an attacker intercepts the SIM card to gain control over the UPI account. - **The absence of SIM change detection (SIM-SWAP-001)** and the lack of new device or first-time beneficiary flags (APP-002) indicate that the account's usual security parameters have not been altered, which is unusual for legitimate transactions of this magnitude. - **Contradictory Signals** - The MuleHunter score (0.180) is below the mule threshold (>0.75), suggesting that the account does not exhibit typical mule account behaviors. This lack of corroboration from the MuleHunter model weakens the suspicion of money muling activities, focusing the investigation primarily on the SIM swap risk.
- Recommended Action and Confidence Level** - **Recommended Action:** Initiate a multi-factor authentication (MFA) challenge for the account holder to verify their identity. This could include sending a one-time password (OTP) to the registered mobile number or email, or prompting the account holder to answer security questions. Additionally, monitor the account for any further suspicious activity and consider temporarily restricting high-value transactions until the account's security can be confirmed. - **Confidence Level:** 0.92. The high confidence level is based on the strong evidence from the VEL-001 rule, which indicates a significant deviation from the account's normal behavior. The behavioral risk score further supports the suspicion of a SIM swap takeover attempt. While the MuleHunter score does not corroborate money muling activities, the overall pattern of evidence strongly suggests that immediate action is necessary to secure the account and prevent potential fraud.

Additional Context

Investigation Findings

The transaction was flagged because it triggered two high-risk rules: 'SIM-SWAP-001' and 'VEL-001', which are associated with the 'SIM_SWAP_UPI_TAKEOVER' pattern. The behavioral risk score is also high, indicating a potential SIM swap takeover attempt, where an attacker may have gained unauthorized access to the customer's UPI account by intercepting the SIM card.

Outcomes, trends and model impact

WHAT YOU'RE SEEING

- Detection rate & auto-handled KPIs
- Geographic & time-of-day fraud distribution
- Live fraud intelligence by channel & typology
- AI escalation impact tracked over time

WHY IT MATTERS Leadership sees the platform's effectiveness and where to tune it.



Every agent action is on the record

WHAT YOU'RE SEEING

- Append-only audit of every tool call & decision
- Each row ties to case, actor, action & event
- An on-screen assistant answers over live cases
- RAG over current cases + the knowledge base

WHY IT MATTERS *Full traceability plus an analyst copilot, grounded in real case data.*

- Command Center
- Investigation
- Analytics
- Agent Audit**
- Learning Loop
- Help & Guide

Agent Audit Trail

143 Total
76 Awaiting
75 Critical
0 SAR
23 Escalated
67 Resolved
Analyst

#	TIMESTAMP	CASE ID	ACTOR	ACTION	DETAIL
1	24 Jun 00:40	CASE-20260623-D7FF903F	output_node	Investigation_complete	confidence=0.92 score=0.87 ops_event=False
2	24 Jun 00:40	CASE-20260623-D7FF903F	knowledge_update_node	knowledge_updated	clarity=CLEAR_FRAUD doc_id=None learning=Extreme ...
3	24 Jun 00:40	CASE-20260623-D7FF903F	compliance_node	compliance_check	ncrp_filed=False mule_reported=False action=block ty...
4	24 Jun 00:40	CASE-20260623-D7FF903F	notification_node	notification_dispatched	action=freeze_account sent=False channel=upi
5	24 Jun 00:40	CASE-20260623-D7FF903F	case_management	case_opened	type=FraudCase
6	24 Jun 00:40	ALERT-20260623-P0010	decision_node	hiti_decision	hiti=False auto_action=freeze_account reason=BLOCK...
7	24 Jun 00:40	ALERT-20260623-P0010	upi_fraud_node	upi_specialist_analysis	confirmed_rules=1/5 confidence=0.92 recommendatio...
8	24 Jun 00:39	ALERT-20260623-P0010	context_enrichment_node	context_enriched	risk=HIGH typologies=[SIM_SWAP_TAKEOVER] runbo...
9	24 Jun 00:39	ALERT-20260623-P0010	orchestrator_node	alert_received	channel=UPI action=block score=0.87
10	24 Jun 00:39	CASE-20260623-9FF3ED0B	output_node	Investigation_complete	
11	24 Jun 00:39	CASE-20260623-9FF3ED0B	knowledge_update_node	knowledge_updated	
12	24 Jun 00:39	CASE-20260623-9FF3ED0B	compliance_node	compliance_check	
13	24 Jun 00:39	CASE-20260623-9FF3ED0B	notification_node	notification_dispatched	
14	24 Jun 00:39	CASE-20260623-9FF3ED0B	case_management	case_opened	
15	24 Jun 00:39	ALERT-20260623-P0009	decision_node	hiti_decision	
16	24 Jun 00:39	ALERT-20260623-P0009	upi_fraud_node	upi_specialist_analysis	
17	24 Jun 00:39	ALERT-20260623-P0009	context_enrichment_node	context_enriched	
18	24 Jun 00:39	ALERT-20260623-P0009	orchestrator_node	alert_received	
19	24 Jun 00:39	CASE-20260623-E7F67B4D	output_node	Investigation_complete	
20	24 Jun 00:39	CASE-20260623-E7F67B4D	knowledge_update_node	knowledge_updated	
21	24 Jun 00:39	CASE-20260623-E7F67B4D	compliance_node	compliance_check	
22	24 Jun 00:39	CASE-20260623-E7F67B4D	notification_node	notification_dispatched	
23	24 Jun 00:39	CASE-20260623-E7F67B4D	case_management	case_opened	
24	24 Jun 00:39	ALERT-20260623-P0008	decision_node	hiti_decision	hiti=False auto_action=freeze_account reason=Bl...
25	24 Jun 00:39	ALERT-20260623-P0008	upi_fraud_node	upi_specialist_analysis	confirmed_rules=1/5 confidence=0.86 recommen...

Connected

help me with ALERT-20260623-P0010

ASSISTANT
ALERT-20260623-P0010 is a **blocked** transaction (action=block) that was automatically stopped by the system (hiti=False). It is not linked to any fraud case, SAR, or pattern cluster in the current data set. No further analyst action is required unless you have additional context.

YOU
What was the agent action?

ASSISTANT
The agent action for ALERT-20260623-P0010 was **BLOCK**.

Ask about cases, decisions, SARs...

Send

The platform proposes its own new rules

WHAT YOU'RE SEEING

- Resolved cases clustered into emerging patterns
- Each gap becomes a PENDING Flink rule proposal
- An analyst must approve before anything deploys
- Approved rules ship as code via a pull request

WHY IT MATTERS *The system learns and suggests — humans still approve every rule change.*

FraudOps
INDIA PAYMENTS

Command Center
Investigation 76
Analytics
Agent Audit
Learning Loop
Help & Guide

Learning Loop
AI-generated rule proposals and emerging fraud pattern clusters. Approve proposals to deploy new Flink rules.

RULE PROPOSALS 0 Refresh

No proposals yet — the agent will surface them as it processes cases.

EMERGING PATTERNS 10 Run Now

UPI < ₹1,000 per txn 10 cases

SIM-Port Account Takeover
10 accounts were compromised following a SIM port attack. Once the attacker controlled the mobile number, they bypassed OTP authentication across banking apps and made transfers (under ₹1,000 each). All 10 cases triggered the same set of detection rules (bc7d3b), suggesting a common attack pattern.
bc7d3b
First seen: 2026-06-23 19:06 Latest: 2026-06-23 19:10 728756d1 Investigate 10 cases →

CARD ₹1,000 – ₹10,000 per txn **NOVEL** 4 cases

Novel / Unclassified Pattern
4 CARD transactions (₹1,000–₹10,000 each) share an identical pattern but don't match any known fraud type in the system's rule library. No existing detection rule matches this signature (334c4a triggered but produced no typology match). This could represent a new fraud method. Review the cases for common attributes and consider raising a rule proposal if the pattern holds.
334c4a
First seen: 2026-06-12 00:58 Latest: 2026-06-20 08:58

UPI < ₹1,000 per txn 15 cases

ACCOUNT TAKEOVER UPI
15 UPI cases showing account takeover upi behaviour triggered the same set of detection rules (cf1082), suggesting a common attack pattern.
cf1082
First seen: 2026-06-08 21:58 Latest: 2026-06-21 11:58

TRANSFER ₹1,000 – ₹10,000 per txn 15 cases

Transaction Structuring (Below CTR)
15 TRANSFER transactions were deliberately sized to stay just below the ₹10 lakh reporting threshold (₹1,000–₹10,000 per txn).
First seen: 2026-06-12 00:58 Latest: 2026-06-20 08:58

Novel pattern — Novel / Unclassified Pattern
4 CARD transactions (₹1,000–₹10,000 each) share an identical pattern but don't match any known fraud type in the system's rule library. No existing detection rule matches this signature (334c4a triggered but produced no typology match). This could represent a new fraud method. Review the cases for common attributes and consider raising a rule proposal if the pattern holds.

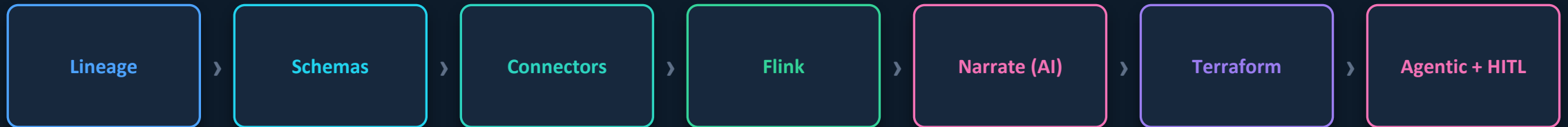
Payment channel	CARD
Transaction amount range	₹1,000 – ₹10,000 per txn
Cases in this group	4 cases
Detection rules triggered	334c4a
First case seen	2026-06-12 00:58
Most recent case	2026-06-20 08:58

Recommended action: Flag for analyst review. Run pattern gap detection. Consider new Flink rule proposal.

● Live · 10:26

FROM RAW PAYMENT TO GOVERNED DECISION

One platform, end to end



PII-Safe

Identifiers hashed at source; raw PII never touches Kafka.

Deterministic Core

Rules, CEP and a 40/40/20 ensemble decide — replayable.

Governed AI

Gateway + Flink connections; agents assist, never decide.

IaC + Open Lakehouse

Terraform everything; Tableflow → Iceberg, zero ETL.

Confluent Kafka + Flink · Schema Registry · Tableflow → Apache Iceberg · HashiCorp Terraform · Governed Agentic AI · on Microsoft Azure